

4(a)	<i>Mycobacterium tuberculosis</i> ; must be spelled correctly	1
4(b)	1 (infected), person, coughs / breathes (out) / talks / sneezes / AW, <u>and</u> (uninfected) person, inhales / inspires / breathes in ; 2 <u>aerosol infection</u> / <u>aerosol transmission</u> / <u>droplet infection</u> / <u>droplet transmission</u> <i>must be a term</i> or organism / pathogen / bacteria / <i>M. tuberculosis</i>, in, airborne droplets / droplets <u>in</u> air ; A contaminated airborne droplets / contaminated droplets in air A ref. to airborne droplets from an infected person transmits the pathogen I infected droplets / infected airborne droplets / droplets of air / air droplets <i>max 1 if another mode of transmission <u>also</u> given as a choice</i> <i>max 1 if correct description of Mycobacterium bovis given e.g. <u>drinking contaminated (unpasteurised) milk</u></i> or <u>eating contaminated meat (from cattle)</u>	2
4(c)	antibiotic ; I examples I antibacterial	1
4(d)	label line and label to: bronchus ; bronchiole ; blood vessel ; R if blood vessel is stated as a capillary	3

Question	Answer	Marks
4(e)	any five from: <i>gas exchange and overall health to max 4</i> 1 fewer, bronchioles / alveoli / alveolar capillaries or granuloma / described, causes, damage / lesions / airway lumen to decrease / AW ; A takes up lung space I blocks airways 2 consequence ; e.g. less air enters alveoli / reduced surface area (for gas exchange) <i>context is fewer alveoli (fewer capillaries)</i> fewer red blood cells / less haemoglobin, to take up oxygen (<i>granuloma</i>) causes (partial) air blockage / prevents alveolar function / AW suggestion that cells in the granuloma take up oxygen 3 many immune system cells / dead cells / debris / AW, hinders gas exchange ; <i>this is not the same as mp8</i> 4 less oxygen, reaches tissues / to the (rest of) body (in circulation) ; 5 consequence ; e.g. tiredness / fatigue / less mobile increases blood pressure / strain on heart rapid breathing / causes breathlessness A difficulty breathing I affects breathing <i>prevent development TB elsewhere in body to max 4</i> 6 infected (lung) cells / pathogen, prevented from entering, circulation / bloodstream ; 7 granuloma acts to, confine / localise / isolate, pathogen ; A description of cells within granuloma acting as the barrier 8 many / surrounded by / AW, cells of the immune system / named ; I white blood cells e.g. lymphocytes / plasma cells / phagocytes / macrophages / neutrophils / multinucleated giant cells <i>note: many macrophages in a giant cell is not mp7</i> 9 other detail of immune response stimulated ; e.g. antigen presentation antibody formation (against <i>M. tuberculosis</i>) phagocytosis of, infected cells / pathogen cytokine release 10 AVP ; e.g multinucleated giant cells, more effective as a large cell / increase chance of detecting pathogen result of immune response prevents, population increase / bacteria reproducing / AW another detail of immune response	5